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[A2-ST-NAME]: Karaca, Osman
[A2-Y-TARGET]: Website Users for “ www.ofka.tr”

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MULTI-CONTEXT MODEL METRICS (From Sections 3 & 4):

- Baseline Multiple Linear Regression R-Squared: [0.1619]
- Algorithmic Random Forest Regressor R-Squared: [0.8709]
- Maximum Context-Driven Prediction Deviation: [18.719170758284164]

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Section 1: The Multi-Context Architecture & Hypothesis

1. A. The Contextual Dimensions:

- The Core Performance Stream

Meta Business Suite, a social media management tool, collects OFKA’s digital marketing activities. These variables represent the actions and elements of Meta users. The performance variables include Instagram Views, Instagram Reach, Facebook link clicks, Instagram profile visits, Instagram follows, Facebook Views, Facebook interactions, Facebook page visits, and viewers.

- Surrounding Environment Stream

The Surrounding environment stream is dated, such as day, month, and quarter. These variables represent background conditions that may influence customer behavior.

- Data Integration Process

There are two main data channels: Google Analytics integrated into WooCommerce and Meta’s channel. 11 different data sets come from Meta; only one set comes from Google Analytics in WooCommerce. Also, there are 4 context variables. All these data were joined to form unified multi-context data sets containing 113 observations and 16 variables.

2. B. The Behavioral Hypotheses:

- Hypothesis A

One Hypothesis claims that a high level of social media engagement will increase link clicks and the number of active users visiting www.ofka.tr.

- Hypothesis B

Another hypothesis suggests that the relationship between website visitors and social media activity might not be linear. Website visits may be higher with a low level of engagement or vice versa. Therefore, a random Forest model might perform better than traditional multiple linear regression.

Section 2: The Feature Inventory & Accessibility Critique

Variable Label	Data Metric Type	The accessibility Critique
Active_Users	Discrete Ratio	This is the Y value that the model will try to predict. It is essential to increase website traffic. Active users are potential customers, and after pulling them into the website, we can see where they get stuck.
IG_Views	Discrete Ratio	It is important to understand if these people were aware and searching the company and how these views are explained to potential customers.
IG_Interactions	Discrete Ratio	The number of likes or reactions, saves, comments, shares, and replies on your content, including ads. Content can include formats such as posts, stories, reels, and more (Meta, 2026). This is important for understanding which reactions are most effective at turning people into active users.
IG_Link_Clicks	Discrete Ratio	Click, swipe, or tap on a link that leads to destinations such as WhatsApp or a website. It captures direct movement from Meta to the Website.
IG_Profile_Visits	Discrete Ratio	It is reliably collected from Meta, which indicates people intentionally come and look at the profile.
IG_Follows	Discrete Ratio	It is a convenient proxy because easily accessible through the Meta Business Suite. While follower growth is a positive indicator, it doesn't directly correlate to active users on the website.
FB_Views	Discrete Ratio	Convenient proxy that is automatically collected by the Mate business suite and easily accessible. These view fluctuations indicate something; even though views are high, they do not translate to active users on the website.
FB_Interactions	Discrete Ratio	It is a high-fidelity signal because it captures engagements through likes, comments, and shares. This provides stronger evidence of customer interest than views alone.
FB_Link_Clicks	Discrete Ratio	It is a high-fidelity metric because it directly measures customer movement from Facebook to OFKA's website. This metric closely reflects behavioral intent and is more valuable than awareness-based measures.
FB_Visits	Discrete Ratio	Like IG_Visits, FB_Visits is a medium-fidelity signal because it indicates intentional visits to the Facebook page.
FB_Follows	Discrete Ratio	It is a convenient proxy that is easily accessible and automatically tracked by Meta. Followers do not equate to website visits or active users.
DayOfWeek	Ordinal	Missing Context Recovery: Customers' behavior might change at the day of the week.
Month	Ordinal	Missing Context Recovery: Customer behavior can change based on the season, such as Christmas or Ramadan.
Quarter	Ordinal	Missing Context Recovery: Represents broader seasonal and business cycle effects.

Weekend	Binary Nominal	Missing Context Recovery: Try to understand the weekend's effect on active users at OFKA.
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Section 3: The Baseline Linear Audit (The Straight-Line Worldview)

Using Python in Visual Studio Code and the scikit-learn Machine learning library, a standard multiple linear regression (MLR) model was developed to predict website Active users using OFKA's Combined Variables. The final data set contained 113 daily observations. The model's baseline achieved an R^2 of 0.1619. This R^2 indicated that the model explained approximately 16.19% of the variation in website traffic.

The Linear Regression Equation:

$$(Y)Active_Users =$$

$$6.68 - 0.0055(IG_Views) - 0.0395(IG_Interactions) + 0.1247(IG_Profile_Visits) - 0.2902(IG_Follows) + 0.0524(FB_Views) - 0.0994(FB_Interactions) - 1.5008(FB_Link_Clicks) - 0.0887(FB_Visits) - 0.6302(DayOfWeek) - 1.2168(Month) + 1.4350(Quarter) + 1.4542(Weekend)$$

The Linearity Critique:

The assumption of a linear link between social media activity and website traffic appears unrealistic for OFKA's digital marketing environment. Every Variable has a fixed impact on the Actual users of the website, which means for example one Facebook view impacts 0.05 on the active users number, yet Instagram interactions have a negative impact. However, customer behavior can be influenced by a combination of factors instead of fixed actions.

The limitation is shown in the model. The multiple linear regression model explains only 16.19 % of the variables. In the other world, 83.81 percent of the variation in website traffic remains unexplained under a straight line framework. This reveals the existence of a hidden threshold, an interaction effect over the combined factors.

Section 4: The Algorithmic Stress Test (The Non-Linear Trigger Audit)

After the multiple linear regression, the model is not reliable because the explanation of variant levels is stuck at 16.19 %; however, after running the random forest regression model, r^2 jumped to 0.8709% which indicates the model explains most of the variants in website traffic.

$$\Delta = \text{PredictedRF} - \text{PredictedMLR}$$

$$Abs_Delta = |\Delta|$$

In this calculation, Delta is converted to its absolute value because it represents the magnitude of disagreement between Multiple Linear Regression (MLR) and Random Forest (RF) models rather than the direction of disagreement. However, to determine whether Random Forest performs better than Multilinear regression, the original Delta must be examined.

The results claim that the Random Forest Model is a more accurate representation of OFKA's website traffic pattern than the multiple Linear Regression model and is more explainable than MLR. The reason is that MLR assumes fixed variables and doesn't account for the combined function mentioned in the sections above.

The Tipping Point Audit

Hidden Traffic Sweet Spot Detected (January 26, 2026):

A major disagreement occurred on Jan 26, 2026. On this date, the MLR model prediction was lower than that of the random forest regression. MLR was 7.723, while random forest was 25.95, and the actual number was 36 active users. This indicates the Random forest model, which is the context-aware worldview, is explained and predicted higher than MLR and achieved a closer prediction for OFKA's website active user.

Table 1: Variables Name converted to Xn for Space limitation.

Date	Y1	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	MLR	RF	Delta	Abs_Delta
2026-01-26	36	8	0	0	4	0	1	0	0	2	0	0	1	1	0	7.23	25.95	18.72	18.72

Traffic Cliff Detected (January 26, 2026):

Multi-Linear Regression assumed there must have been 4+ people, and the Random Forest Model predicted 1.4 people (1 person there can't be a 0.4 individual). Models without real-life function and flexibility, like multilinear regression, can be dangerous to interpret for humans. The behavior was supposed to be better because the prediction was supposed to be better, but the fact is that only one person was an active user that day.

Table 2: Variables Name converted to Xn for Space limitation.

Date	Y1	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12	X13	X14	MLR	RF	Delta	Abs_Delta
2026-05-12	1	40	0	0	4	1	30	0	0	0	0	1	5	2	0	4.4	1.44	-2.96	2.96

Section 5: The "Misdirected Rigor" Verdict

Finally, the definitive empirical results show that OFKA's website traffic environment is absolutely not linear, and the multiple linear regression (MLR) model fails to explain most of the variance, with an R^2 of only 16.19%. By contrast, the random forest model, which accounts for real-world conditions and complexity, explains 87% of the variance in Ofka's website traffic.

The large performance gap shows that customer behavior is influenced by complex interactions, not just social media engagement. Contextual variables like month or season (Ramadan in Turkey, New Year's Eve, etc.) have a significant impact on customer behavior and website traffic. These complex relationships and combined factors are explainable with the random forest model.

Continuing to manage OFKA's digital marketing performance using averages would result in a type III error because leadership would be optimizing the relationship with solely the engagement number. There are action items they need to look at the research. Why even our engagement is good but fails to get traffic, and why thousands of people watch ads but are not responsible, there are questions awaiting to be answered.

Appendix:

GitHub Repository: https://github.com/osmanfatihkaracausa-svg/Mgt8040_Data_Mining_Osman

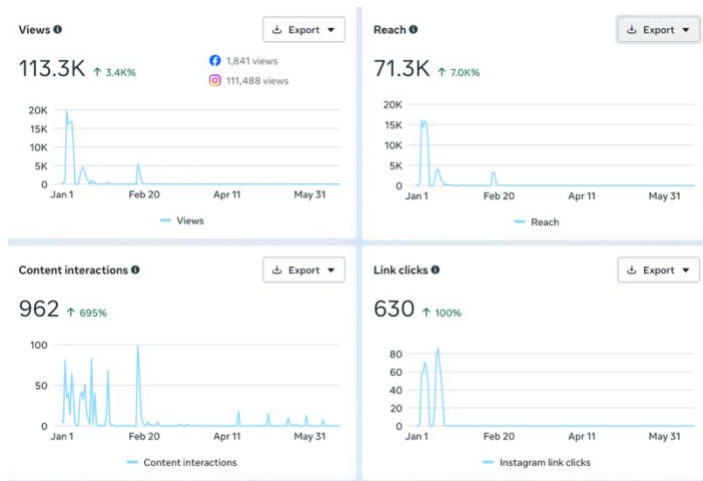


Figure 1: Sourced from Meta Business Suit- Data owner is Ofka.tr

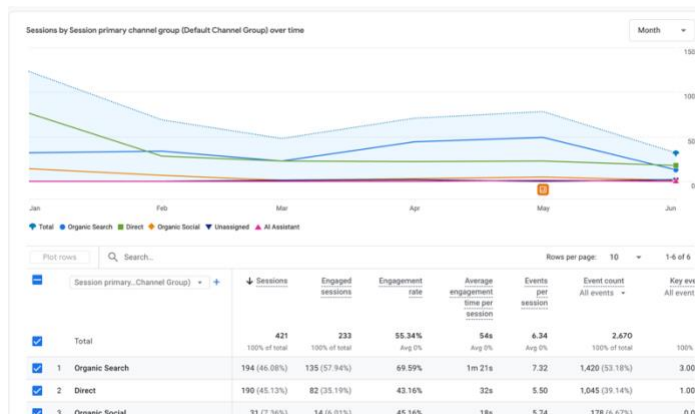


Figure 2 : Sourced from Meta Business Suit- Data owner is ofka.tr